

Introduction to Machine Learning

Course Project and Resources

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This document outlines the examination requirements and provides some useful resources for the course.

Contents

1	Overview	2
1.1	Important Dates	2
1.2	Deliverables	2
1.3	Grading	3
2	Course Project	3
2.1	Tasks	3
2.2	Comments	4
3	Term Paper	4
4	Datasets	5
5	Anaconda, Python, and Notebooks	5
5.1	Anaconda	5
5.2	JupyterLab and Jupyter Notebook	5
5.3	Resources	6
5.4	Requirements	6

1 Overview

The Course Project applies and compares machine learning techniques to a self-chosen problem class of your interest. The project will be conducted in teams of 3-4 members.

1.1 Important Dates

Task	Deliverable	Due Date
1	Project Proposal	04/17/2026
2	Project Milestone	05/13/2026
3	Final submission	07/03/2026

1.2 Deliverables

Submissions: All submissions should be uploaded to the course site on ELO by one member of a group on behalf of all other members of the group. Each submission should include the names of all group members.

Project Proposal: The project proposal (pdf) should be one paragraph. It should contain the following information:

- Tentative title of the term paper.
- Names of all group members.
- Problem you will investigate and why the problem is interesting.
- Short description of the data you will use.

Project Milestone: Your project milestone will be a commented notebook for Task 2 of the course project (see Section [2.1](#)). The notebook should have the following structure:

- Title
- Authors
- Imports and Global Settings
- Data
- Experiments
- Discussion

Final Submission: The final submission consists of

- Term paper (pdf)
- Notebooks of Tasks 3 and 4 for verification of the results (see Section 2.1)
- A document detailing the contributions of each team member to the project

Further information on the term paper can be found in Section 3.

1.3 Grading

Only the term paper will be graded. All other deliverables are necessary for verification purposes and to ensure that you are progressing as expected.

2 Course Project

2.1 Tasks

The practical element of the course project consists of the following four tasks:

1. Choose a problem: The problem should consist of at least three datasets for classification from a specific domain. The restriction is that the data must be tabular data.
2. Compare baseline classifiers
 - Logistic Regression
 - k-nearest Neighbors (self-study)
 - Naive Bayes (self-study)
3. Compare
 - best performing baseline
 - SVM
 - MLP
 - Random Forests
 - Gradient Boosting
4. Model analysis of the best performing method.

You are encouraged and welcome to add other promising methods from the literature not mentioned here for comparison.

2.2 Comments

Self Study. To enhance the models, you are encouraged to explore additional techniques not covered in the lectures (e.g. dimensionality reduction, feature selection). This will require independent study. For instance, the k-nearest neighbor classifier and the Naive Bayes classifier are not included in the course material. You are expected to learn these techniques on your own to successfully complete Task 2.

Model Analysis. In this task, you will systematically investigate a specific component or behavior of your best-performing model to understand its impact on overall performance. This may include, for example, a *Sensitivity Analysis* (varying specific hyperparameters) or a *Feature Importance Study* (identifying which variables drive predictions). The goal is to provide a deeper insight into why the model performs the way it does.

Experimental Methodology You are expected to demonstrate your ability to conduct experiments in a methodologically correct and fair manner. This includes: choice of appropriate metrics, careful hyperparameter selection, sound evaluation, and fair comparison.

3 Term Paper

The paper should not exceed 12 pages from introduction to conclusion. It can be written in German or in English. It should follow the structure below:

- Title, Authors (1-2 lines)
- Abstract (1 paragraph)
- Introduction (1 page)
- Related Work (1 page)
- Methods (2-3 pages)
- Experiments (2-3 pages)
- Conclusion (1-2 paragraphs)
- References
- Appendix (optional)

The course site offers a LaTeX template along with instructions on how to write a term paper. For those new to LaTeX or looking for a convenient way to write their paper, [OverLeaf](#) is highly recommended. Overleaf is an online LaTeX editor that requires no installation and provides a wide range of templates to get you started.

Note that LaTeX is recommended for this course but its use is not mandatory. You are free to use your favorite word processor.

4 Datasets

The following list includes some standard dataset repositories:

- [UCI Machine Learning Repository](#)
- [Kaggle](#)
- [Papers With Code](#)
- [Google Dataset Search](#)

5 Anaconda, Python, and Notebooks

This course uses interactive programming with Jupyter Notebooks. This section outlines the basic steps needed to create a functional system environment.

5.1 Anaconda

Anaconda is a distribution of the Python and R programming languages particularly suited for data science and machine learning. You can choose between Miniconda and Anaconda. Choose Anaconda if you are new to Python or prefer convenience over manual installation and configuration. Consult the [documentation](#) for further information.

5.2 JupyterLab and Jupyter Notebook

Jupyter Notebook is a widely used interactive tool in the data science and machine learning community for programming Python. JupyterLab is an advanced version of Jupyter Notebook with a more convenient user interface. To install and launch JupyterLab, proceed as follows:

1. Download and install [Anaconda](#)
2. After installation, start the Anaconda-Navigator app
3. Optionally create a new environment in the Environment screen of Anaconda-Navigator or use the default environment base.
4. Install JupyterLab in the Home screen of the Anaconda-Navigator app. Ensure that you have selected your desired environment.
5. Launch JupyterLab.

5.3 Resources

- [The Python Tutorial](#)
- [Anaconda-Navigator documentation](#)
- [JupyterLab documentation](#)
- [Jupyter Notebook documentation](#)

5.4 Requirements

For the course we require the following packages:

- `numpy` [\[>\]](#)
- `scikit-learn` [\[>\]](#)
- `matplotlib` [\[>\]](#)
- `pandas` [\[>\]](#)
- `scipy` (optional) [\[>\]](#)

Most packages come preinstalled in Anaconda. Check the index on Anaconda's environment screen to see which packages are missing and install them.